Kasper GPS prototype review — 12 Oct

Oct 12, 2026 · @afzl

View as is broken in your copy because of a demo-bar bug that the H4 branch already fixes but `main` does not have yet. On the latest build, every user sees exactly the right assets. What lets the demo down is the data and a few privacy leaks: machines look parked at demo time, alert times are wrong, the tracking page has no map, and Site Users and renters see things they shouldn't.

Why View as looks broken

View as itself works. The problem is the demo bar in the version you're running. The bar is a 36 px strip set to scroll sideways, and that setting also clips anything that drops down from it. So clicking View as or Scenarios opens the list *inside* the strip. You see only its top edge (the "Search users…" box), and scrolling inside the strip pushes the buttons out of sight. That is exactly your screenshot.

Image — Your screen: the View as and Scenarios lists trapped inside the demo bar

Version	Where	Date	View as
Your local copy	`Gpsarena\kaspergps`, branch `h2-correctness-fixes`, plus 7 unsaved edits	9 Oct	Broken (clipped)
`main` on GitHub	H1–H3 merged	10 Oct	Broken (clipped)
`h4-spec-features` (PR #14, not merged)	Latest work	12 Oct, 10:15	Works: a real dropdown with a role filter, users grouped by company, and hardware badges

Image — The H4 build: View as opens as a dropdown with a role filter and company groups

To switch users today, either:

1. Sign out and sign in with any seeded email. Any password works (for example `omar@alnoor.ae`, `khalid@emiratesearth.ae`, `lina@marina.ae`, `sara@kasper.ae`). This works on every version.

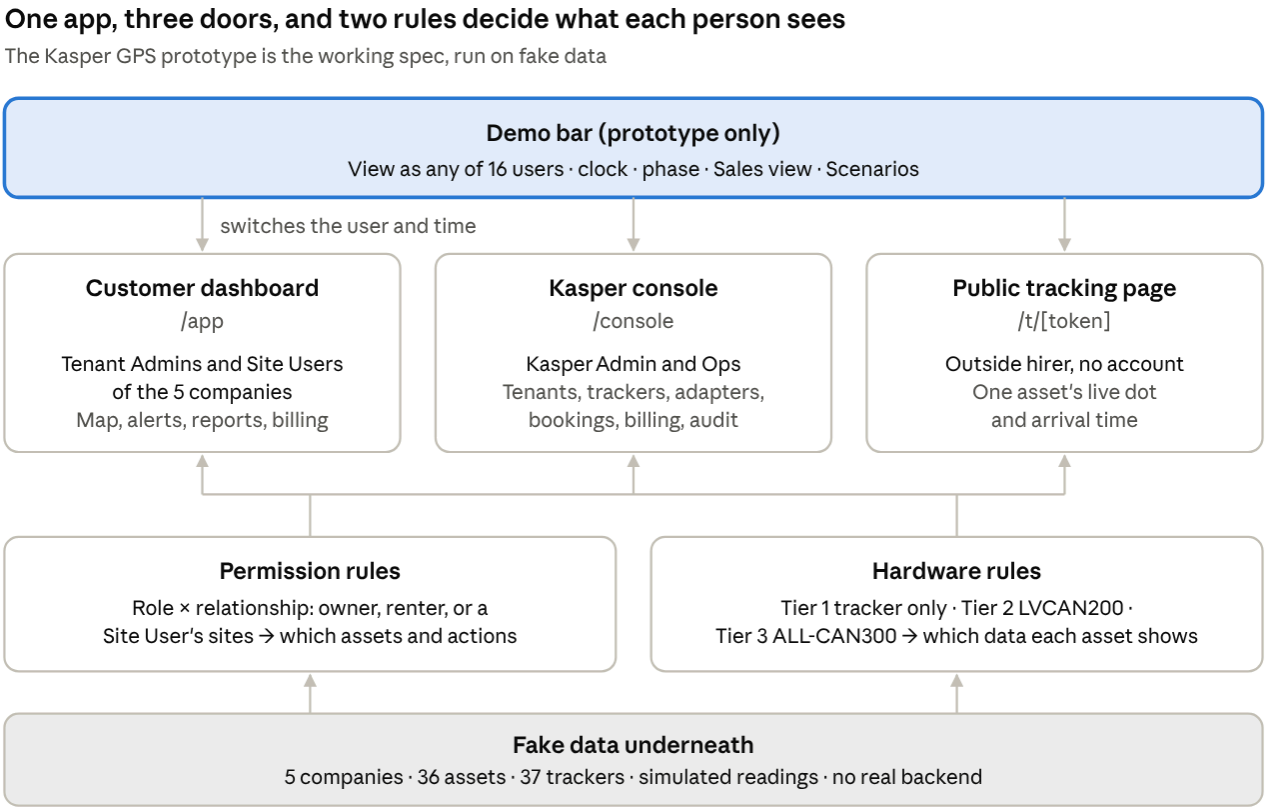
2. Run the latest build in a new folder. Your current folder and its unsaved edits stay untouched:

```
cd C:\Users\HP\Desktop\antigravity\Gpsarena
git clone -b h4-spec-features https://github.com/afsalali1238/kaspergps
kaspergps-latest
cd kaspergps-latest
npm ci
npm run dev
```

Then open `http://localhost:3000/sign-in` and use View as. The permanent fix is to merge PR #14 into `main`.

What we're building

A clickable prototype of the Kasper GPS dashboard that serves as the working spec. The founder, ops, interns and dev team can switch to any user and see exactly what that user would see before production is built. The permission and hardware rules must be exact. Everything underneath is fake.



Kasper GPS prototype · 3 surfaces, 2 rule sets, fake data

- **Who it's for:** the people deciding the rules (founder, ops, interns), and the dev team who will build production from it.

- **What it is not:** the product. There are no real trackers, backend, email or payments. Production swaps the data layer (`src/server/db.ts`) for a database and sign-in for real auth. It also swaps the simulator for FMC130 data arriving over TCP/UDP (Traccar, Kasper's own listener, or AWS IoT Core).
- **The cast:** 5 companies with different hardware, 16 users across 4 roles, and an outside hirer. Al Noor has no CAN adapters, Emirates is mostly Tier 3, and Palm only rents.
- **The clock:** the app's time is fixed at Tue 6 Oct, 10:00, so every demo starts from the same moment. The demo bar moves it.

Where it stands

The core rules are right. What's left is the simulated data, a few privacy leaks in the UI, and screens that fall short of the spec. Checked on the latest H4 build.

Area	Verdict	What I saw
Who sees which assets	Works	All 15 customer users see exactly the assets the spec lists. Priya on EX-04 and Fatima on EX-11 get "Asset not found". Karim is refused at sign-in.
Kasper console	Works	Sara gets all 12 sections. Ravi (Ops) has no Billing, Kasper team, Audit log or Onboarding.
Hardware tiers	Partly	Omar never sees fuel, RPM or Certificates. Sales view shows locked "Needs ALL-CAN300 (Tier 3)" cards. EX-04 shows ECU hours of 8,420 h. The Engine & fuel tab is still thin.
Renter view	Partly	Lina's EX-04 page is right: "Rented from Emirates Earthmovers … History starts 4 Oct 08:00", with only Run report. But renters see the owner's yard names.
Simulated data	Broken	Shifts run 11:00–22:00 instead of 07:00–18:00, so at the demo time every machine is parked.
Alerts	Broken	The times in an alert's text and its "Since" line disagree, and one alert is dated in the future.

Area	Verdict	What I saw
Public tracking page	Broken	No map and no dot. Only the ETA line shows.
Settings	Partly	Site Users get admin buttons, and customers see tracker IMEIs.
Maintenance, Cost & ROI, Billing	Works	Basis labels, "Dummy rates", the Draft tag, and working Pay and Record payment flows.
Features panel (the reviewer's explainer)	Partly	A raw list of permissions with no reasons. The hardware section is empty.
Stability	Works	130+ page visits with no crashes or console errors, and no sideways scroll at 390 px on the home screens.
Automated browser tests	Not run	Not run since H1, per the 11 Oct handoff.

Each user: what they should see and what they see

Every user's asset list matches the spec. The differences are in what surrounds it: Site Users get admin controls, renters see the owner's yards, and the data makes working machines look parked.

User	Role · company	Spec says they see	They see now	Verdict
Sara Haddad	Kasper Admin	Whole console, every tenant, audit	All 12 console sections; can open any asset	Works
Ravi Menon	Kasper Ops	Trackers, assets, bookings; no admin, billing or audit	8 console sections; no Billing, Kasper team, Audit log or Onboarding	Works
Omar Saleh	Tenant Admin · Al Noor (no CAN)	10 Tier 1 assets; never fuel or engine data	10 assets (Live 7, Idle 1, Stale 1, Offline 1); no Certificates; locked cards in Sales view. FB-12 should be driving to Al Habtoor but shows 0 km today	Partly

User	Role · company	Spec says they see	They see now	Verdict
Hessa Al Mansoori	Tenant Admin · Al Noor	Same as Omar	Same as Omar	Partly
Khalid Rahman	Tenant Admin · Emirates (Tier 3)	12 own assets; ECU hours, fuel, faults, certificates	12 assets; EX-04 ECU 8,420 h; good Maintenance and Cost & ROI. EX-04 ("Working") shows engine off, 0 rpm, 0% load	Partly
Priya Nair	Tenant Admin · Gulf Lift	11 own assets + rented BD-02; EX-04 = not found	Exactly that	Works
Mark D'Souza	Site User · Gulf Lift, Al Qusais	Al Qusais assets + BD-02, view only	Right 12 assets, but Settings gives him Invite, Deactivate, Add asset, Retire and tracker IMEIs	Partly
Lina Aziz	Tenant Admin · Marina (renter)	PU-51, PU-52, VN-01 + rented EX-04 and CR-02	Exactly that; EX-04 renter view correct	Works
Ahmed Yousef	Site User · Dubai Hills	EX-04 + PU-51	Right 2 assets; Settings admin buttons	Partly
John Mathew	Site User · Business Bay + JVC	CR-02, PU-52, VN-01	Right 3 assets; Settings admin buttons	Partly
Anil Kumar	Site User · JVC Villas	VN-01	Right asset; Settings admin buttons	Partly
Karim Old	Site User · deactivated	Refused at sign-in	"Your account is no longer active."	Works
Sam Invite	Site User · invited	Signs in; Business Bay	CR-02 and PU-52; Settings admin buttons	Partly

User	Role · company	Spec says they see	They see now	Verdict
Fatima Noor	Tenant Admin · Palm (renter only)	WL-06, TH-01, GN-01; EX-11 gone	Exactly that; EX-11 "Asset not found". Gets an Owned switch and empty Maintenance and Cost pages, though Palm owns nothing	Partly
Rashid Ali	Site User · Palm Crescent	WL-06	Right asset; owner's yard shown; Settings admin buttons	Partly
Deepa Shah	Site User · Dubai South + Palm Crescent	TH-01, GN-01, WL-06; fuel-drop alert, view only	Right 3 assets, two-site filter, no Acknowledge. Rows show the owners' yards	Partly
Outside hirer	No account	One live dot for FB-12	ETA line, but no map and no dot	Broken

Problems found, by priority

Eleven problems. The first three spoil any live demo, the next three break the permission rules the prototype exists to prove, and the rest are screens short of the spec.

Spoils the demo

1. **Machines are parked at demo time.** The simulator starts each day at midnight UTC, which is 04:00 in Dubai, so work shifts start four hours late: 07:00–18:00 becomes 11:00–22:00. The app's clock is fixed at Tue 6 Oct, 10:00, an hour before any shift starts. EX-04 ("Working" in the spec) shows ignition off, 0 rpm and 0% load. FB-12 should be driving to Al Habtoor but shows 0 km today. FB-12, WT-07 and EX-04 all log one trip a day, 11:00–22:00, about 10.7 km at a top speed of 25 km/h: a truck and an excavator look the same.

Image — EX-04 at 10:00: the "working" excavator reads 0 rpm and 0% load

Image — FB-12 trips: the same 11:00–22:00, 25 km/h trip every day

2. **Alert times contradict the alert text.** TP-23 reads "Power cut at 12:10" but is listed "Since 5 Oct 20:10", and "Offline since 14:32" is listed "Since 5 Oct 19:19". The 14:32 is the spec's example wording, copied in as fixed text. GN-01 reads "Fuel dropped 18% at 02:10" but is listed "Since 6 Oct 12:10", two hours after the app's "now".
3. **The public tracking page has no map.** The hirer sees a "Live map" label, the asset name and the ETA line, but no map and no dot (scenario S14).

Image — The outside hirer's page: no map

Breaks the permission rules

4. **Site Users get admin controls.** Every Site User (Mark, Ahmed, John, Anil, Sam, Rashid, Deepa) has Settings in the menu, with Invite user and Deactivate. Mark's also offers Add site, Delete, Add asset and Retire. The system does block the action (Mark's Deactivate on Priya changed nothing), but it does so silently. The spec says these buttons are hidden.
5. **Customers see tracker IMEIs.** Settings → Assets lists each tracker's 15-digit IMEI. The spec says customers never see trackers, IMEIs or SIMs. The same table shows every status as "unknown" while the map says Live or Idle.

Image — Mark, a Site User: Add asset, Retire on every row, and tracker IMEIs

6. **Renters see the owner's yards.** Deepa's rented WL-06, GN-01 and TH-01 are listed at Ras Al Khor Yard, Al Quoz Yard and Al Qusais Yard (Emirates' and Gulf Lift's yards), not at her sites. Her alert filter lists those yards too. She also gets an Owned / Rented In switch, though Palm owns nothing.

Image — Deepa's map: rented assets listed at the owners' yards

Short of the spec

7. **The Features panel doesn't explain the user.** The spec has it group every permission with its scope and the reason for each ×, list the hardware tiers and features with reasons, and show "right now" counts. It shows a raw list ("Asset ViewTelemetry", "Grant EndEarly") and an empty Features section. This is the panel a reviewer uses to understand each user.

Image — The Features panel for Omar

8. **The Engine & fuel tab is thin.** It has no engine hours, no fuel chart with refuel and drop markers, no fuel used per day, and no AdBlue or fault codes. It labels values "CAN" where the spec says "ECU". On Tier 3, the Overview labels "Today's ignition-on time" as ECU; it should say Estimated.
9. **The map layout buries the list.** The map takes the full width, and the filters and asset list sit below it, so at 1440 px the first asset starts below the fold. The spec puts the list beside the map. Asset codes wrap inside their badges ("FB-" over "12").

Image — Omar's map: filters and list below the map

10. **On a phone, the demo bar takes 4 rows** (about 145 px), and the header loses the company name.

Image — Phone, 390 px: the demo bar wraps to four rows

11. **Small things:**
    - "0 min ago(hover for details)" has no space and reads as a note to the developer.
    - "Palm Contracting ·1 Sep" is missing a space.
    - Lina's invoice reads "6 OctOverdue Overdue".
    - The utilisation days are out of order (1 Oct, 2 Oct, 29 Sep, 3 Oct, 30 Sep).
    - Fuel reads 53% on every Tier 3 asset checked.
    - The Scenarios menu shows 3 of its 50 scenarios at a time.
    - Palm, which owns nothing, gets empty Maintenance and Cost & ROI pages.

Changes needed, in order

Do these in this order. The first two make the demo usable, and items 3–6 fix the rules the founder and dev team will judge it on. Each item can be handed to the agent as written. The spec (`BUILD_PROMPT.md`) still wins on any detail.

1. **Merge PR #14 (`h4-spec-features`) into `main`.** This fixes View as. Then everyone runs a fresh clone of `main`, not the 9 Oct copy.

2. **Start the simulator's day at Dubai midnight.** `src/server/telemetry/simulator.ts` computes day starts as `Math.floor(now / 86400000) * 86400000` (UTC). Use `clock.startOfDubaiDay` instead. Then check at 10:00 on 6 Oct:
   - EX-04 is working (rpm 1,200–2,100, load 35–85%);
   - FB-12 is driving to Al Habtoor and arrives about 90 minutes later;
   - trucks drive between sites at 20–90 km/h, and plant stays within about 300 m of its site.
3. **Compute every alert's time and text from its scenario time.** Remove the copied example strings ("12:10", "14:32"). No alert may be dated after the app's "now".
4. **Put the map back on the public tracking page.** It should show one marker for FB-12, with the ETA line under it (S14, S29).
5. **Close the Settings leaks:**
   - Site Users don't get Settings (`users.manage` and `sites.manage` are ✕ for them);
   - any button a user can't use is hidden, not refused silently;
   - remove the Tracker (IMEI) column from every customer screen;
   - the asset table uses the same status as the map.
6. **Show rented assets at the renter's site.**
   - Rows, the map and alert lines use the booking's renter site, never the owner's yard.
   - Site filters list only the viewer's own sites.
   - Hide Owned / Rented In unless the user has both.
7. **Rebuild the Features panel to spec §10.2:**
   - permissions grouped, each with ✓ or ✕, its scope in plain words, and the reason for each ✕ (from `capability-reasons.ts`);
   - the hardware tiers in view, and every feature with ✓ or ✕ and its reason;
   - "Right now" counts.
8. **Complete the Engine & fuel tab to spec §11.3:**
   - engine hours with their source;
   - a 7-day fuel chart with refuel ▲ and drop ▼ markers, and fuel used per day;
   - AdBlue, or "Not measured";
   - fault codes with plain descriptions.
   - Source labels are ECU, ECU · partial or Estimated only. Today's ignition-on time is always Estimated.
9. **Map layout:**
   - at desktop width, the list sits beside the map with the filters above both;
   - on phones, use Map | List tabs;
   - the asset-code badge never wraps.

10. **Phone demo bar:** one row. View as and the clock stay visible, and everything else goes behind one "Demo" menu.
11. **Copy fixes** from problem 11, plus sorting utilisation days by date.
12. **Run the full Playwright suite on desktop and phone, and save the screenshots (H6, H7).** Nothing on the screen has been browser-tested since H1. Done means a green CI run.

How this was checked: the H4 branch at commit `659e738` (12 Oct, 10:15), built and run as a production build, then signed in as all 16 users plus the outside hirer: 130+ page visits at 1440 px and 390 px. Map tiles were stubbed because the test machine can't reach OpenStreetMap. Everything else is the real app.